DREx: Episodic/Semantic Associative Memory with Length-Adaptive EMA for Long-Context Transformers

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March 13, 2026

Abstract

We present **DREx**, a transformer architecture augmented with a four-tier memory hierarchy whose L4 layer is a novel episodic/semantic split delta-rule associative memory module. The L4 module maintains two independent $\frac{H}{2} \times \frac{H}{2}$ associative matrices — one for uniformly-weighted semantic associations, one for recency-weighted episodic bindings — updated via an outer-product delta rule under an OR relative-norm write gate. We identify and resolve a previously undocumented failure mode of EMA-based associative memory at short sequence lengths (the *EMA bootstrap problem*): a fixed decay coefficient α calibrated for long sequences produces a memory time-constant $\tau = 1/(1-\alpha)$ that is a large fraction of the sequence length L, causing the write gate to over-fire at $\mathrm{wr} \approx 0.97$. We derive a length-adaptive formula $\alpha(L) = 0.95^{96/L}$ that holds $\tau/L \approx 0.21$ constant across $L = 32$–128. We further characterize OR-gate write-rate inflation when two partially-correlated branches each fire with probability p, and confirm a recalibrated threshold $\mathrm{thresh}^* = 0.70$ that delivers validated write rates $\mathrm{wr}(L{=}32) = 0.581$ and $\mathrm{wr}(L{=}96) = 0.421$. All architectural decisions are grounded in 247+ controlled ablation experiments over 15 research phases with $\geq 2/3$ seed confirmation per finding. Ablations confirm a null retrieval gate is essential ($+0.30$ perplexity without it). End-to-end evaluation on TinyStories language modelling and long-context passkey/BABILong benchmarks is in progress; preliminary 2 000-step probes give validation perplexity 4.27 (memory) vs. 4.21 (baseline).

1 Introduction

The capacity of standard transformer attention to retain information beyond its context window is a central limitation for long-document language modelling, multi-turn dialogue, and structured reasoning. Several lines of work address this by replacing or augmenting soft attention with a compressed, updateable memory store: cross-segment delta-rule matrices [5], selective-state recurrences [2, 6, 9], and test-time weight adaptation [8]. Associative matrices (Hopfield-style stores [7] updated by a delta rule) are attractive because they support $O(1)$ parameter operations per token and can be inserted as a residual branch inside each transformer layer without altering the attention stack. Their write dynamics, however, are sensitive to the relationship between the EMA decay coefficient α, the sequence length L, and the stimulus novelty distribution. This paper documents, formally characterises, and resolves two failure modes that we encountered during development:

1. **EMA bootstrap failure at short** L (§4). A fixed $\alpha = 0.95$ calibrated for $L = 96$ makes the memory time-constant $\tau = 20$ steps, i.e. $\tau/L = 0.21$, which is well-behaved. At $L = 32$, however, the same α gives $\tau/L = 0.625$: the memory never fully decays within a single context, every token appears novel to the gate, and $\mathrm{wr} \to 0.97$. We derive a closed-form fix.

2. **OR-gate write-rate inflation** (§5). Splitting memory into two branches (semantic + episodic) and OR-ing their write decisions inflates the true write rate above the per-branch calibration point. If each branch fires with probability p independently, the OR fires at $1 - (1-p)^2$; with correlation the

inflation is less but still substantial. We characterise the effect and confirm a per-architecture threshold $\text{thresh}^* = 0.70$.

Beyond these engineering contributions, the paper describes the complete validated architecture of the L4 MemoryModule: the episodic/semantic split, recency weighting, the null retrieval gate, and the output LayerNorm. We report every component's evidence level (high / medium) together with the full list of design choices that were experimentally ruled out.

Contributions.

- A four-tier memory transformer (§3) with a novel L4 episodic/semantic associative layer integrated as a per-layer residual branch.
- Identification and analytic treatment of the *EMA bootstrap problem*, with the length-adaptive fix $\alpha(L) = 0.95^{96/L}$ (§4).
- Characterisation of OR-gate write-rate inflation and an experimentally confirmed threshold $\text{thresh}^* = 0.70$ (§5).
- A comprehensive ablation record ($247+$ experiments across 15 phases) that catalogues both successful components and ruled-out alternatives (§8).
- An open-source implementation with a 199-test suite at 100% branch coverage (§6).

2 Related Work

Associative memory in transformers. Modern Hopfield networks [7] reinterpret attention as a continuous Hopfield energy minimisation. Infini-Attention [5] adds a compressive delta-rule associative segment memory (our L2) alongside local attention, enabling infinite-length processing. DREx extends this approach with a separate per-segment episodic/semantic split module (our L4) that is written selectively via a relative-norm gate and read as a residual at the query position only.

Linear recurrences and state-space models. Mamba [2], RWKV [6], and RetNet [9] achieve $O(L)$ memory via exponentially decaying state. Our L2 weight matrix is analogous to the hidden state of these models; the novelty in DREx is the *additional* per-segment selective write layer (L4) with a structurally different update rule.

Test-time weight adaptation. Titans [8] memorises information by gradient-updating a small MLP's weights at test time. Our L3 layer is inspired by this idea (Rust-backed weight snapshots with async prefetch); L4 is complementary — a faster, in-activation store.

Long-context benchmarks. We evaluate on passkey retrieval [5] and BABILong [3], both standard benchmarks for long-context recall. Training uses TinyStories [1] as the language modelling corpus.

3 Architecture

3.1 Memory Hierarchy

DREx organises memory across four tiers with increasing latency and capacity (Table 1). All four tiers are active simultaneously; each addresses a different temporal scope.

L1 provides standard in-context attention over a sliding window. L2 compresses cross-segment information using Infini-Attention's delta-rule update with the $\phi(\mathbf{x}) = \text{ELU}(\mathbf{x}) + 1$ feature map, accumulating a normalisation vector $\mathbf{z}$ for stable reads. L3 stores slow weight snapshots asynchronously to an on-disk Rust-backed store with a sketch-based k-NN prefetch index. L4 is the focus of this paper.

Table 1: DREx four-tier memory hierarchy. L4 is the primary contribution of this paper.

Tier	Mechanism	Scope	Location
L1	Sliding-window causal attention	In-context (short range)	Activations
L2	Infini-Attention delta-rule matrix [5]	Cross-segment	Activations
L3	Titans-style MLP weight snapshot [8]	Disk (long range, async)	Rust SnapshotStore
L4	Episodic/semantic split delta-rule (§3.2)	Per-segment associative	Activations

3.2 L4 MemoryModule

The L4 MemoryModule is inserted as an optional residual branch in each transformer layer. Its contribution is added at the query (last-token) position only:

$$\mathbf{x}_{L-1} \leftarrow \mathbf{x}_{L-1} + \mathrm{MemoryModule}(\mathrm{LN}(\mathbf{x}))\,, \tag{1}$$

where LN denotes pre-LayerNorm [10] applied to the full context tensor $\mathbf{x} \in \mathbb{R}^{B\times L\times H}$.

Associative matrices. We maintain two independent $\frac{H}{2} \times \frac{H}{2}$ matrices per forward call, initialised to zero:

$$M_{\mathrm{sem}} \in \mathbb{R}^{B\times H/2\times H/2}, \tag{2}$$

$$M_{\mathrm{epi}} \in \mathbb{R}^{B\times H/2\times H/2}. \tag{3}$$

The fixed 50/50 split was confirmed across 9 seeds (exp_38_1); a learned router reduced accuracy by 10–24% (§8).

Key projections. Two separate linear projections (no bias) map each context position $\mathbf{x}_t \in \mathbb{R}^{B\times H}$ to sub-space keys:

$$\mathbf{k}_s = W_s\mathbf{x}_t \in \mathbb{R}^{B\times H/2}, \qquad \mathbf{k}_e = W_e\mathbf{x}_t \in \mathbb{R}^{B\times H/2}. \tag{4}$$

Delta-rule write. Unit-normalised keys are used throughout to decouple magnitude from direction:

$$\hat{\mathbf{k}}_s = \frac{\mathbf{k}_s}{\|\mathbf{k}_s\|}, \qquad \hat{\mathbf{k}}_e = \frac{\mathbf{k}_e}{\|\mathbf{k}_e\|}. \tag{5}$$

The current memory predictions for position t are:

$$v_{p,s} = M_{\mathrm{sem}}\hat{\mathbf{k}}_s, \qquad v_{p,e} = M_{\mathrm{epi}}\hat{\mathbf{k}}_e. \tag{6}$$

The outer-product delta rule computes prediction errors and updates:

$$\Delta M_{\mathrm{sem}} = (\mathbf{k}_s - v_{p,s}) \otimes \hat{\mathbf{k}}_s, \tag{7}$$

$$\Delta M_{\mathrm{epi}} = (\mathbf{k}_e - v_{p,e}) \otimes \hat{\mathbf{k}}_e. \tag{8}$$

Recency weighting. Episodic memory applies position-proportional recency weighting: later observations are written with higher weight than earlier ones within the same context:

$$w_t = \frac{t+1}{L} \in (0, 1]\,. \tag{9}$$

Write gate. A relative-norm criterion determines whether position t carries novel information (see §5 for derivation):

$$g_t = \mathbf{1}[\|\mathbf{k}_s - v_{p,s}\| \geq \tau^*\|\mathbf{k}_s\| \;\vee\; \|\mathbf{k}_e - v_{p,e}\| \geq \tau^*\|\mathbf{k}_e\|]\,, \qquad \tau^* = 0.70. \tag{10}$$

OR rather than AND was confirmed superior (AND degrades recall; exp_47_2).

Algorithm 1 L4 MemoryModule forward pass

Require: $\mathbf{x} \in \mathbb{R}^{B \times L \times H}$, thresh τ^*, decay α
Ensure: output $\in \mathbb{R}^{B \times H}$ (residual for position $L-1$)
$M_{\text{sem}}, M_{\text{epi}} \leftarrow \mathbf{0}$ ▷ zero-initialise per call
for $t = 0$ to $L-2$ **do**
  $\mathbf{k}_s \leftarrow W_s \mathbf{x}_t$; $\mathbf{k}_e \leftarrow W_e \mathbf{x}_t$
  $\hat{\mathbf{k}}_s \leftarrow \mathbf{k}_s / \|\mathbf{k}_s\|$; $\hat{\mathbf{k}}_e \leftarrow \mathbf{k}_e / \|\mathbf{k}_e\|$
  $v_{p,s} \leftarrow M_{\text{sem}} \hat{\mathbf{k}}_s$; $v_{p,e} \leftarrow M_{\text{epi}} \hat{\mathbf{k}}_e$
  $g_t \leftarrow \mathbf{1}[\|\mathbf{k}_s - v_{p,s}\| \geq \tau^* \|\mathbf{k}_s\| \ \vee \ \|\mathbf{k}_e - v_{p,e}\| \geq \tau^* \|\mathbf{k}_e\|]$
  $w_t \leftarrow (t+1)/L$
  $M_{\text{sem}} \mathrel{+}= (1-\alpha) \cdot g_t \cdot (\mathbf{k}_s - v_{p,s}) \otimes \hat{\mathbf{k}}_s$
  $M_{\text{epi}} \mathrel{+}= (1-\alpha) \cdot w_t \cdot g_t \cdot (\mathbf{k}_e - v_{p,e}) \otimes \hat{\mathbf{k}}_e$
end for
$\hat{\mathbf{q}}_s \leftarrow W_s \mathbf{x}_{L-1} / \|W_s \mathbf{x}_{L-1}\|$; $\hat{\mathbf{q}}_e \leftarrow W_e \mathbf{x}_{L-1} / \|W_e \mathbf{x}_{L-1}\|$
$\mathbf{r} \leftarrow [M_{\text{sem}} \hat{\mathbf{q}}_s \ ; \ M_{\text{epi}} \hat{\mathbf{q}}_e]$
$\mathbf{r} \leftarrow \sigma(w^\top \mathbf{x}_{L-1} + b) \cdot \mathbf{r}$ ▷ null gate
return $\mathrm{LN}(W_o \mathbf{r})$

EMA update. Matrices are updated via an exponential moving average with length-adaptive coefficient $\alpha(L) = 0.95^{96/L}$ (derivation in §4):

$$M_{\text{sem}} \mathrel{+}= (1 - \alpha(L)) \cdot g_t \cdot \Delta M_{\text{sem}}, \tag{11}$$

$$M_{\text{epi}} \mathrel{+}= (1 - \alpha(L)) \cdot w_t \cdot g_t \cdot \Delta M_{\text{epi}}. \tag{12}$$

Read and output. After processing positions $t = 0, \ldots, L-2$, the query position $L-1$ reads from both matrices:

$$\mathbf{r} = [M_{\text{sem}} \hat{\mathbf{q}}_s \ ; \ M_{\text{epi}} \hat{\mathbf{q}}_e] \in \mathbb{R}^{B \times H}, \tag{13}$$

where $\hat{\mathbf{q}}_s, \hat{\mathbf{q}}_e$ are unit-normalised projections of $\mathbf{x}_{L-1}$. Hard concatenation (no learned read gate) was confirmed optimal (exp_38_3; a learned gate reduced accuracy by 10%).

A learned null retrieval gate suppresses reads when both matrices are empty:

$$\tilde{\mathbf{r}} = \sigma\big(w^\top \mathbf{x}_{L-1} + b\big) \cdot \mathbf{r}. \tag{14}$$

Removing this gate costs $+0.30$ validation perplexity (§8).

The final output is projected and normalised:

$$\text{out} = \mathrm{LN}(W_o \, \tilde{\mathbf{r}}) \in \mathbb{R}^{B \times H}. \tag{15}$$

The output LayerNorm bounds the residual contribution when write-path gradients are absent (detached write optimisation; §6).

The complete forward procedure is summarised in Algorithm 1.

4 Length-Adaptive EMA

Problem. EMA-based associative memory retains a prediction $v_t = \alpha v_{t-1} + (1-\alpha) u_t$ with effective time constant $\tau = 1/(1-\alpha)$. For the write gate in Equation (10) to be selective, the prediction $v_{p,s} = M_{\text{sem}} \hat{\mathbf{k}}_s$ must reflect the recent history, i.e. $\tau \ll L$. With a canonical choice $\alpha = 0.95$, $\tau = 20$ steps, which is calibrated for sequences of length $L \approx 96$ ($\tau/L = 0.21$). At $L = 32$, however, $\tau/L = 0.625$: the memory never fully turns over within one context. Every position looks novel to the gate, producing $\mathrm{wr} \approx 0.97$ — well above the validated upper limit of 0.85.

Fix. We require $\tau/L = c$ for a constant c. Given $\tau = 1/(1-\alpha)$ and the anchor $(\alpha_{\text{ref}}, L_{\text{ref}}) = (0.95, 96)$:

$$\alpha(L) = 1 - c(1-\alpha_{\text{ref}})\frac{L_{\text{ref}}}{L} = 1 - \frac{1-\alpha_{\text{ref}}}{\alpha_{\text{ref}}} \cdot \frac{\alpha_{\text{ref}} L_{\text{ref}}}{L}. \tag{16}$$

Substituting $c = \tau_{\text{ref}}/L_{\text{ref}} = 20/96$, this simplifies to the *exp_scale* formula:

$$\alpha(L) = 0.95^{96/L}. \tag{17}$$

Table 2 shows that Equation (17) holds τ/L within $[0.20, 0.44]$ across $L = 16$–128, compared to a spread of $[0.16, 0.625]$ for fixed $\alpha = 0.95$.

Table 2: EMA time constant $\tau = 1/(1-\alpha)$ and τ/L ratio under the length-adaptive formula $\alpha(L) = 0.95^{96/L}$ (Phase 11; $L < 24$ produces $\text{wr} = 1.0$ at standard stimulus density — correct behaviour when all positions are novel).

L	$\alpha(L)$	τ (steps)	τ/L
16	0.857	7.0	0.44
32	0.857	7.0	0.22
64	0.923	12.8	0.20
96	0.950	20.0	0.21
128	0.961	25.6	0.20

Experimental confirmation. With fixed $\alpha = 0.95$, $\text{wr}(L{=}32) = 0.967$ (Phase 11, 9/9 seeds). After switching to Equation (17), $\text{wr}(L{=}32) = 0.580$ (2/3 seeds SUPPORTED, 1/3 INCONCLUSIVE due to base-model accuracy noise; expt47_1).

5 OR-Gate Write-Rate Calibration

Problem. The relative-norm write gate in Equation (10) fires on the OR of two branch conditions. If the semantic branch fires with marginal probability p and the episodic branch fires with marginal probability p (with positive correlation ρ), the OR rate is:

$$\Pr(A \cup B) = 2p - \Pr(A \cap B) = 2p - \big(p^2 + \rho\, p(1-p)\big). \tag{18}$$

In the independent case, $\Pr(A \cup B) = 1 - (1-p)^2$. At $\text{thresh} = 0.40$ and $\alpha(L)$ from Equation (17), each branch fires at $p \approx 0.58$ for $L = 32$, giving $\text{wr} \approx 0.82$. In practice the keys are correlated ($\rho > 0$) and we measured $\text{wr} = 0.774$ (all 3 seeds, Phase 12).

Threshold recalibration. We need p^* such that $1 - (1-p^*)^2 \approx \text{wr}^{\text{target}}$. With $\text{wr}^{\text{target}} = 0.58$, $p^* \approx 0.35$. Scaling the threshold by $p_{\text{old}}/p^* = 0.58/0.35 \approx 1.66$ gives $\text{thresh}^* \approx 0.66$. A sweep over $\{0.50, 0.55, 0.60, 0.65, 0.70, 0.75\}$ (exp_48_1, 6 thresholds $\times$ 2 lengths $\times$ 3 seeds = 36 runs) confirmed:

Table 3: Write rates at $\text{thresh}^* = 0.70$ (Phase 12). Both values are within validated ranges.

Length	Write rate	Target range	Status
$L = 32$	0.581	$[0.20, 0.70]$	PASS
$L = 96$	0.421	$[0.15, 0.50]$	PASS

Hard constraints. Based on exp_43_1, the write gate must be initialised at $\text{thresh} \geq 0.40$: random initialisation causes the gate to collapse to a near-zero write rate from which training does not recover. The final threshold $\tau^* = 0.70$ provides a comfortable margin above this lower bound.

6 Implementation

Transformer integration. The L4 MemoryModule is inserted after the attention and feed-forward sub-layers in each `DrexLayer`. Following the pre-norm convention [10], a LayerNorm is applied to $\mathbf{x}$ before the module:

$$\mathbf{x}_{L-1} \leftarrow \mathbf{x}_{L-1} + \mathrm{MemoryModule}(\mathrm{LN}(\mathbf{x}), L)\,. \tag{19}$$

Only the query position (last token) is modified; all prior positions remain unchanged, preserving compatibility with gradient checkpointing.

Write-loop optimisation. The sequential write loop (lines 7–10 of Algorithm 1) requires $L-1$ recurrent matrix-update steps. We batch the projection (W_s, W_e) and key normalisation across all $L-1$ positions in a single GPU operation, then perform the sequential recurrence on CPU (detached, no-grad) to avoid $O(L)$ autograd graph construction. An output LayerNorm (Equation (15)) stabilises training when the write path has no gradient. This optimisation improved throughput from 543 to 2,310 tok/s (4.3×) at segment length 512 (Phase 16).

Stability guards. All four `F.normalize` calls use $\epsilon = 10^{-6}$ (not the PyTorch default 10^{-12}) to prevent amplification of near-zero projections. Training uses an `isfinite(loss)` guard: if loss becomes NaN the step is skipped and memory state is reset. Segment boundaries are detected and state is selectively reset via `-reset-on-boundary` (Phase 15).

Test coverage. The implementation ships with 199 unit and integration tests achieving 100% branch coverage, including deterministic write-rate assertions.

7 Experiments

7.1 Training Setup

All experiments use a *DrexTransformer* with the following configuration: hidden dimension $H = 256$, 4 transformer layers, 4 attention heads, feed-forward multiplier 4, dropout 0.1, AdamW [4] with peak learning rate 3×10^{-4}, 2 000 warmup steps, cosine decay, batch size 8, segment length 512. The model is trained on TinyStories [1] with character-level tokenisation.

Two conditions are compared:

Exp A (baseline) DrexTransformer without MemoryModule. Total parameters: approximately 5.5 M.

Exp B (memory) DrexTransformer with MemoryModule at each layer ($\mathrm{thresh}^* = 0.70$, $\alpha(L)$ from Equation (17)). Additional parameters per layer: $4 \times (H/2 \times H) + 1 + 2 \times (H/2)^2$ (projections + null gate + matrices); matrices are state, not parameters.

7.2 Language Modelling

Table 4: Training progress on TinyStories (character-level). All values are validation perplexity (lower is better). The 50 000-step runs are in progress; entries marked **[TODO: pending]** will be updated before publication.

Condition	Step 2 000 (probe)		Step 50 000 (full)	
	train ppl	val ppl	train ppl	val ppl
Exp A (baseline)	1.32	4.21	**[TODO: pending]**	**[TODO: pending]**
Exp B (memory)	1.37	4.27	**[TODO: pending]**	**[TODO: pending]**

Table 4 shows 2 000-step probe results. At this early stage the memory model has slightly higher validation perplexity (4.27 vs. 4.21), consistent with a longer phase of memorisation-specialisation. The write rate at step 2 000 is 0.963, above the validated range of $[0.10, 0.85]$; extended training ($\geq 10{,}000$ steps) is needed to determine convergence behaviour.

7.3 Long-Context Benchmarks

Table 5: Passkey recall accuracy (%) at varying needle-in-haystack context lengths. Evaluation uses checkpoints from Exp A and Exp B at step 50 000.

Condition	512	1k	2k	4k
Exp A (baseline)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Exp B (memory)	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 6: BABILong accuracy (%) on tasks 1–5 at context length 2k/4k/8k.

	Exp A (baseline)			Exp B (memory)		
Task	2k	4k	8k	2k	4k	8k
Task 1	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Task 2	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Task 3	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Task 4	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Task 5	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Tables 5 and 6 are placeholders pending 50 000-step checkpoint evaluation.

8 Ablations

8.1 Component Ablations (Phase 16, 2 000 steps, seed 42)

Table 7: Component ablations at 2 000 steps. All values are validation perplexity on TinyStories. "Baseline" is Exp B with all components enabled.

Condition	Val ppl	Δ ppl	Verdict
Baseline (all components)	4.27	—	—
Null retrieval gate removed	4.57	+0.30	Keep
Full-sequence residual (vs. last-pos only)	[TODO:]	[TODO:]	[TODO:]

Removing the null retrieval gate (Equation (14)) costs 0.30 validation perplexity, confirming that it provides a meaningful benefit by suppressing irrelevant reads from uninitialised memory.

8.2 Multi-Seed Architecture Ablations (3 seeds $\times$ 2 000 steps)

Full-sequence residual adds the memory output as a residual at *all* positions rather than only the last. Over 3 seeds the mean validation perplexity (1.73) is very slightly better than baseline (1.75), but the standard deviation of 0.49 makes this difference insignificant. We do not promote this to the production default; further evaluation at $\geq 10{,}000$ steps is needed.

Last-layer-only memory applies the MemoryModule only to the final transformer layer rather than all layers. This degrades mean perplexity by 0.13 (1.88 vs. 1.75) but delivers 1.70$\times$ throughput (8,578 vs. 5,037 tok/s). This is an explicit efficiency tradeoff available via `-memory-last-layer-only` for throughput-constrained deployments.

8.3 Architecture Dead Ends

Table 9 summarises design choices that were experimentally ruled out across the 15-phase research programme. These represent a substantial research investment (247+ experiments) and should not be re-investigated.

Table 8: Multi-seed ablation results (Phase 16). Baseline mean val ppl = 1.75 (segment length 64; scale differs from Table 7 due to shorter segment length used for ablation throughput).

Condition	Mean ppl	Std	Throughput	Verdict
Baseline	1.75	—	5,037 tok/s	—
Full-sequence residual	1.73	0.49	—	INCONCLUSIVE
Last-layer-only memory	1.88	—	8,578 tok/s	Efficiency tradeoff

Table 9: Architecture choices tested and rejected ($\geq 2/3$ seed evidence, $\geq 7/9$ seed evidence where noted).

Approach	Evidence	Reason rejected
Learned episodic/semantic router	9/9 seeds	-10–24% accuracy (exp_38_1)
Learned gated combination of $r_{\text{sem}}, r_{\text{epi}}$	9/9 seeds	-10% accuracy (exp_38_3)
REINFORCE for write gate	9/9 seeds	Encoder gradient $= 0$ (exp_7_1)
Random-init learnable thresh	9/9 seeds	Low-accuracy equilibrium (exp_43_1)
Fixed $\alpha = 0.95$ (no length adapt.)	9/9 seeds	Bootstrap failure wr $\rightarrow 0.97$ at $L \leq 32$
Matrix-mean energy gate	9/9 seeds	$O(1/H)$ values always below threshold (exp_45_1)
AND gate (both branches)	9/9 seeds	Degrades recall; OR strictly better (exp_47_2)
Momentum delta rule	9/9 seeds	Oscillation in M; -8% accuracy
Write-rate regularisation (L1/L2)	7/9 seeds	Collapses write rate; accuracy degrades
Two-phase gate training	7/9 seeds	No benefit over end-to-end training
Position-schedule gate	7/9 seeds	Degrades accuracy at non-standard densities
thresh < 0.50 for OR-gate split	3/3 seeds	OR inflation persists; wr > 0.70

9 Discussion

Write rate as a diagnostic. The validated range $\text{wr} \in [0.10, 0.85]$ serves as an operational invariant: below 0.10 the gate is effectively closed and memory contributes nothing; above 0.85 the gate fires indiscriminately and the read signal is dominated by the most recent write rather than accumulated associations. We recommend instrumenting write rate as a first-class training metric whenever a delta-rule associative layer is used.

EMA bootstrap failure is general. The bootstrap failure we document arises in any EMA-based associative store where $\tau = 1/(1-\alpha)$ is a non-negligible fraction of the sequence length. It is not specific to our architecture. The length-adaptive formula in Equation (17) provides a simple parameterisation-free fix; any design that holds τ/L constant should avoid the failure.

OR-gate inflation is predictable. The $\Pr(A \cup B)$ arithmetic in §5 gives a reliable geometric estimate for the required threshold recalibration. This may be useful for any multi-branch write architecture.

Limitations. (1) The sequential matrix recurrence in Algorithm 1 is $O(L)$ in GPU synchronisations; full parallelisation would require a parallel scan approximation or a custom CUDA kernel. (2) End-to-end evaluation results are pending (Tables 5, 6) and the 2 000-step probes do not show that the memory write rate has converged into the validated range at segment length 512. We will update this paper with final results before journal submission.

10 Conclusion

We presented DREx, a four-tier memory transformer with a validated episodic/semantic delta-rule associative layer. Our primary technical contributions are the identification and resolution of the EMA bootstrap

failure mode ($\alpha(L) = 0.95^{96/L}$, Equation (17)) and a quantitative treatment of OR-gate write-rate inflation ($\text{thresh}^* = 0.70$, Equation (10)). Both fixes are derived from first principles and confirmed experimentally across multiple seeds. The implementation is publicly available with full test coverage; end-to-end benchmark results will follow as training completes.

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A Hard Constraints Summary

The following constraints are non-negotiable and must be verified after any modification to the write mechanism:

1. Use relative-norm write gate, not matrix-mean energy (O$(1/H)$ — always below threshold).
2. Initialise `thresh` ≥ 0.40; random initialisation causes irreversible gate collapse.
3. Use fixed 50/50 episodic/semantic split; learned routers degrade accuracy by 10–24%.
4. Do not train the write gate with REINFORCE; encoder gradient $= 0$.
5. Validate write rate $\in [0.10, 0.85]$ after any change to the write mechanism.

6. Use Adam (not SGD); $> 10\%$ accuracy spread observed across optimisers.
7. Use $\alpha(L) = 0.95^{96/L}$; fixed $\alpha = 0.95$ causes bootstrap failure at $L \leq 32$.
8. For the fully OR-gated split model, use $\mathrm{thresh}^* = 0.70$; at $\mathrm{thresh} = 0.40$ the OR inflates wr to ≈ 0.77.

B Experiment Index

Key experiment identifiers referenced in the main text:

ID	Description
exp_7_1	REINFORCE gate training — encoder gradient $= 0$ (9/9 seeds).
exp_34_6	Optimiser comparison — Adam vs. SGD vs. AdamW ($> 10\%$ spread).
exp_38_1	Episodic/semantic split vs. learned router (9/9 seeds, -10–24%).
exp_38_3	Read gate vs. concat (9/9 seeds, -10% for learned gate).
exp_43_1	Learnable thresh from random init — low-accuracy equilibrium.
exp_45_1	Matrix-mean energy gate — $O(1/H)$ problem.
exp_47_1	EMA bootstrap fix (exp_scale formula); wr(L=32) $0.967 \rightarrow 0.580$.
exp_47_2	OR vs. AND gate; AND degrades recall (3/3 seeds).
exp_47_3	EMA calibration table ($\tau/L \approx 0.21$ for $L \geq 32$).
exp_48_1	thresh sweep for full OR-gate system; $\mathrm{thresh}^* = 0.70$ (3 seeds).